

- 5.15 **Relate the conditions for fusion to the difficulty of making a practical and economic form of power station**
- 5.16 Demonstrate an understanding that new scientific theories, such as 'cold fusion', are not accepted until they have been validated by the scientific community

Topic 6

Advantages and disadvantages of using radioactive materials

- 6.1 Explain what is meant by background radiation, including how regional variations within the UK are caused in particular by radon gas
- 6.2 Recall the origins of background radiation from Earth and space

Unit P2: Physics for your future

- 6.3 Describe uses of radioactivity, including:
 - a household fire (smoke) alarms
 - b irradiating food
 - c sterilisation of equipment
 - d tracing and gauging thicknesses
 - e diagnosis and treatment of cancer
- 6.4 Describe how the activity of a radioactive source decreases over a period of time
- 6.5 Recall that the unit of activity of a radioactive isotope is the Becquerel, Bq
- 6.6 Recall that the half-life of a radioactive isotope is the time taken for half the undecayed nuclei to decay
- 6.7 Use the concept of half-life to carry out simple calculations on the decay of a radioactive isotope, including graphical representations
- 6.8 *Investigate models which simulate radioactive decay*
- 6.9 Demonstrate an understanding of the dangers of ionising radiation in terms of tissue damage and possible mutations and relate this to the precautions needed
- 6.10 Describe how scientists have changed their ideas of radioactivity over time, including:
 - a the awareness of the hazards associated with radioactive sources
 - b why the scientific ideas change over time
- 6.11 Discuss the long-term possibilities for storage and disposal of nuclear waste
- 6.12 Evaluate the advantages and disadvantages of nuclear power for generating electricity, including the lack of carbon dioxide emissions, risks, public perception, waste disposal and safety issues

Unit P3: Applications of physics

Overview

Content and How Science Works overview

This highly engaging unit builds on the knowledge gained in units P1 and P2 by introducing students to medical physics. Students will learn how physics principles are vital in modern medicine in the way in which they are applied to diagnosis, treatment and storage of medicines.

Practical work in this unit will give students opportunities to plan practical ways to answer scientific questions; devise appropriate methods for the collection of numerical and other data; assess and manage risks when carrying out practical work; collect, process, analyse and interpret primary and secondary data; draw evidence-based conclusions; and evaluate methods of data collection and the quality of the resulting data.

Students will explain ideas and processes using models while studying radiation, radioactive decay, subatomic structure, momentum and kinetic theory. Work on the intensity of radiation, lenses, momentum, kinetic energy, frequency and gases will provide students with opportunities to work quantitatively. They will have opportunities to communicate scientific information using scientific and mathematical conventions and symbols during work on ray diagrams and vision, and when constructing nuclear equations.

Students will consider the role that physics and physicists play in our lives through the study of laser and other treatments for correcting vision, the use of X-rays in medicine, ECGs and pacemakers, and the uses of radioactive materials. They will consider advantages, disadvantages and risks of using radioactive materials, and consider how decisions about their use are made. They will look at how international collaboration is necessary for the building of particle accelerators and how these can lead to new discoveries about the world around us.

In Topic 1 students will learn about the use of radiation and other waves in medical treatment and diagnosis. Students will apply their understanding of lenses to treatments for long and short sightedness.

In Topic 2 students will learn about the production of X-rays and then discuss the risks and advantages of using X-rays for treatment and diagnosis. A brief study of the use of an electrocardiogram (ECG) will enable students to develop a simple understanding of the use of a pacemaker to regulate heart action.

In Topic 3 students will discuss the ethical and social issues relating to the use of radioactive techniques in medical physics. Students will learn in detail about the decay of radio isotopes and the use of beta decay. They will relate their knowledge to medical treatments and diagnosis and the dangers of using radiation.

Unit P3: Applications of physics

In Topic 4 students will investigate circular motion, momentum and conservation of energy. Students will develop an understanding of particle accelerators and their use in medical physics and wider research.

In Topic 5 students will learn simple kinetic theory and the gas laws. They will learn the general gas equation and how to apply this equation to, for example, bottled gases in medicine.

Assessment overview

This unit is externally assessed, through a one hour, 60 mark, tiered written examination, containing six questions.

The examination will contain a mixture of question styles, including objective questions, short answer questions and extended writing questions.

Practical investigations in this unit

Within this unit, students will develop understanding of the process of scientific investigations, including that investigations:

- use hypotheses which are tested
- require assessment and management of risks
- require the collection, presentation, analysis and interpretation of primary and secondary evidence including the use of appropriate technology
- should include review of methodology to assess fitness for purpose
- should include a review of hypotheses in the light of outcomes.

The following specification points are practical investigations that exemplify the scientific process and may appear in the written examination for this unit:

- 1.8 *Investigate variations of image characteristics with objects at different distances from a converging lens*
- 1.18 *Investigate the critical angle for perspex/air or glass/air or water/air boundaries*
- 1.19 *Investigate TIR between different media*
- 4.12 *Investigate factors affecting the height of rebound of bouncing balls*
- 5.7 *Investigate the temperature and volume relationship for a gas*
- 5.9 *Investigate the volume and pressure relationship for a gas*

The following are further suggestions for practical work within this unit:

- *Investigate the relationship between the intensity of radiation and the distance from the source*
- *Investigate the absorption of light by translucent materials in order to simulate X-rays' absorption*

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- *Investigate conservation of energy and momentum during collisions using models to represent particles*
- *Investigate inelastic collisions with the two objects remaining together after the collision and also 'near' elastic collisions*
- *Investigate the temperature and pressure relationship for a gas*

The controlled assessment task (CAT) for the GCSE in Physics will be taken from any of these practical investigations (specification points and further suggested practical investigations). This task will change every year, so future CATs will be chosen from this list.

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Detailed unit content

In this specification bold text refers to higher tier only content. Italic text refers to practical investigations, which students are required to demonstrate an understanding of.

Throughout the unit

- 0.1 Use equations given in this unit, or in a given alternate form
- 0.2 **Use and rearrange equations given in this unit**
- 0.3 Demonstrate an understanding of which units are required in equations

Topic 1

Radiation in treatment and medicine

- 1.1 Demonstrate an understanding of the methods that medical physicists can employ to help doctors solve medical problems, including:
 - a CAT scans
 - b ultrasounds
 - c endoscopes
 - d ionising and non-ionising radiation
- 1.2 Use the word 'radiation' to describe any form of energy originating from a source, including both waves and particles
- 1.3 Demonstrate an understanding that the intensity of radiation will decrease with distance from a source and according to the nature of the medium through which it is travelling
- 1.4 **Use the equation:**
intensity = power of incident radiation / area
$$I = \frac{P}{A}$$
- 1.5 Describe the refraction of light by converging and diverging lenses
- 1.6 Relate the power of a lens to its shape
- 1.7 Use the equation:
power of lens (diopetre, D) = 1/focal length (metre, m)
- 1.8 *Investigate variations of image characteristics with objects at different distances from a converging lens*
- 1.9 **Use the lens equation:**
$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

(f = focal length (m), u = object distance (m), v = image distance (m))
The use of the real is positive sign convention is preferred and will be used in the exam
- 1.10 Identify the following features in a diagram of the eye – cornea, iris, pupil, lens, retina, ciliary muscles
- 1.11 Demonstrate an understanding that light is focused on the retina by the action of the lens and cornea

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- 1.12 Recall that the average adult human eye has a near point at about 25 cm and a far point at infinity
- 1.13 Explain the symptoms and causes of short sight and long sight (students will not be expected to draw scaled ray diagrams, but may be expected to interpret them)
- 1.14 Compare and contrast treatments for short sight and long sight, including the use of:
 - a simple lenses
 - b contact lenses
 - c **laser correction**(combined lens equation is not required; students will not be expected to draw scaled ray diagrams, but may be expected to interpret them)